ALICE L WALTON SCHOOL OF MEDICINE

Antifungals–Drugs to Treat Fungal Infections

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Adapted from: Drugs to Treat Fungal Infections

Listen to this Brick
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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Describe the mechanism of action and mechanism of resistance of antifungal drugs.
2. List the indications for antifungal drugs used in the treatment of fungal infections.
3. Recognize th d with antifungal drugs.
4. Identify the common drug interactions associated with antifungal drugs.

CASE CONNECTION

Eight-year-old PL and her mother are at your office for a scalp problem. "She's missing a patch of hair in the back. I think it's getting bigger. She's been brushing her own hair. I just saw this." Just as PL's mother described it, there is a well-demarcated area with missing hair and scales and small black dots where the hair has broken off. PL tells you, "It gets itchy. I try not to scratch it, but I need to. Some of the kids in my class say they have worms. I don't believe them."

Should PL believe her classmates about the worms? Consider your answer as you read, and we'll revisit PL at the end of the brick.

GO TO CONCLUSION ↓

What Is Antifungal Therapy?

Before diving into this set of antimicrobial drugs, let's review some relevant biology regarding fungi. Like us humans, they are eukaryotic and require an aerobic environment. This makes it challenging to target the fungi without causing significant host toxicity. They also have cell membranes, but they are unique in having ergosterol (Figure 1). Ergosterol is one of the targets of antifungals.

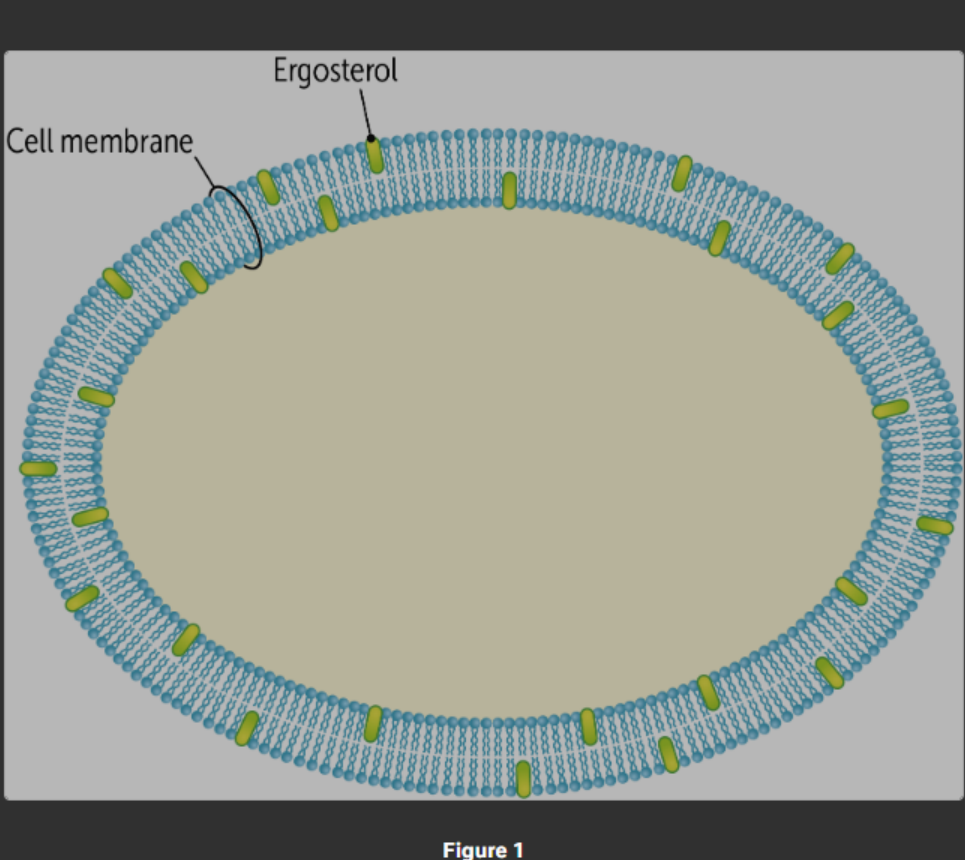


Figure 1
Credit: ©ScholarRx

Fungi have cell walls (surrounding the cell membrane) that are potent antigens to the human immune system. Some fungi have a capsule surrounding the cell wall, particularly *Cryptococcus neoformans* (Figure 2). Antifungal therapy present unique set of challenges in comparison to antibacterial therapy.

- As mentioned above, antifungals can cause significant toxicity since they can also target the host cells.
- Fungal infection ues or avascularized structures such as the hair, skin, and nails.
- Additionally, fungi are also opportunistic. The infection can depend on a variety of host factors, which can impact the prognosis of the disease.

Why is antifungal therapy often more challenging than antibacterial therapy?

What are Antifungal Agents? (LO 1, 3, and 4)

Antifungal agents (AFAs) are drugs that selectively eliminate fungal pathogens from a host with minimal toxicity to said host. They are selectively toxic to fungi because they interact with or inhibit the synthesis of fungal cell walls and ergosterol (which, remember, is only found in fungal cell membranes). Current AFAs include polyenes, azoles, echinocandins, griseofulvin and terbinafine, and flucytosine. We will discuss each drug class next. The description for each drug class describes the mechanism of action, resistance patterns, unique pharmacokinetic properties, and associated ADRs. Figure 1 summarizes the site o sed below.

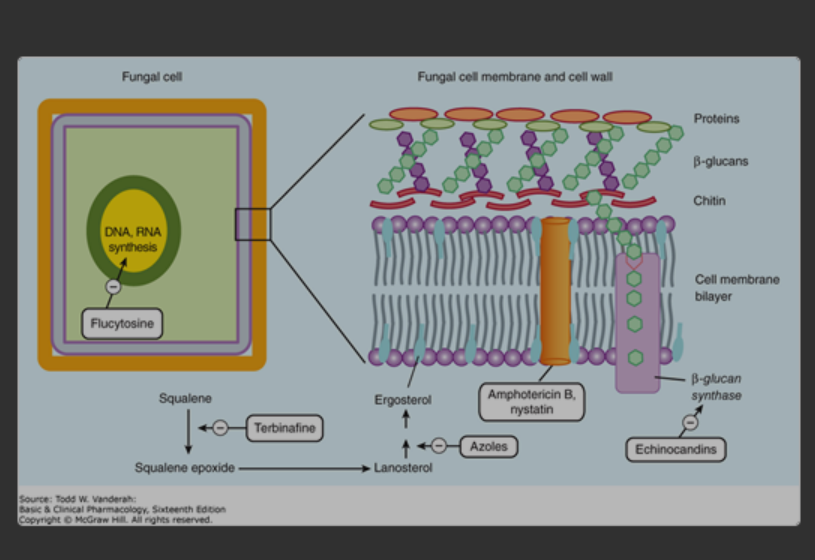


Figure 1: Site of Action for Antifungals

Polyenes

- *Examples:* Amphotericin B
- *Mechanism of action:* Bind to ergosterol and cause formation of pores (holes) in the cell membrane. The pores alter permeability in the fungal cell membrane, eventually leading to content leaking out of the cell causing fungal cell death. Hence the polyenes act as fungicidal.
- *Resistance:* Resistance can occur by decreasing membrane concentration of ergosterol or modification of sterol target, thereby reducing the affinity for the drug.
- *Pharmacokinetics:* Amphotericin B is poorly absorbed from the GI tract and is usually administered via IV as a nonlipid colloidal suspension, a lipid complex, or a liposomal formulation (Note: the addition of lipid decreases the drug's nephr l tissues except the central nervous system (CNS). It is eliminated mainly via slow hepatic metabolism, and a small fraction is excreted in the urine.
- *Adverse drug reactions:* Amphotericin B is nephrotoxic, and renal impairment occurs very frequently. Reversible nephrotoxicity is associated with decreased renal perfusion, while irreversible renal damage, typically from prolonged use, results from renal tubular injury and subsequent dysfunction. Renal toxicity can cause renal tubular acidosis and severe potassium and magnesium wasting. Amphotericin B can also cause nephrogenic diabetes insipidus. Anemia can result since erythropoietin (EPO) is made in the kidney. Intrathecal administration may cause seizures and neurologic damage.

CLINICAL CORRELATION

EPO is a hormone that induces red blood cell production, so if there is not enough EPO made from the kidney, then there are less red blood cells available in the blood, hence the anemia.

Azoles

- *Examples:* Drugs ending with "azole" as a suffix. Ketoconazole, Itraconazole, Fluconazole, Voriconazole, Miconazole, and Clotrimazole are commonly used.
- *Mechanism of Action:* Inhibit the formation of ergosterol by inhibiting 14-α-sterol demethyl lanosterol to ergosterol (inhibi greater affinity for fungal CYP than human. Azoles function as a fungistatic (inhibit the growth of the fungal species)
- *Pharmacokinetics:* For azoles, oral bioavailability is variable since typical gastric acidity is required. Fluconazole and voriconazole are more reliably absorbed via the oral route than are the other azoles. They are distributed to most body tissues, except fluconazole, for which the CNS drug level is very low. All azoles are eliminated hepatically, except fluconazole is renaly eliminated and requires renal dose adjustment.
- *Adverse drug reactions:* check the text. Azoles have a class-wide effect on Liver Function Tests (LFTs) and several significant drug interactions and can increase QT prolongation (ECG changes associated with arrhythmia).
- *Drug-drug interactions:* Several of the azoles are inhibitors of the CYP450 enzyme system (Table 1). Many of the azoles are CYP3A4 inhibitors. Fluconazole, for example, is an inhibitor of the CYP450 enzyme system, increasing the levels of some drugs that are substrates of the CYP2C9 system (e.g., warfarin). Ketoconazole is also an inhibitor of CYP450 enzymes, including the mammalian CYP450 enzymes, thereby disrupting gonadal and adrenal steroid synthesis, which can lead to gynecomastia, decreased libido, and menstrual irregularities.

Table 1 Drug interactions with Azoles (+ = potency of CYP inhibition)

Drug	CYP2C9	CYP2C19	CYP3A4
Fluconazole	+++	+++	++
Ketoconazole	+	+	+++
Itraconazole			+++
Voriconazole			+++

What is the mechanism of action of azoles?

Echinocandins

- *Examples:* Caspofungin, Micafungin, and Anidulafungin ("Fungin" suffix)
- *Mechanism of action:* These large lipopeptide molecules are inhibitors of β-glucan synthesis, which disrupt the fungal cell wall. β-Glucan is a fungal cell wall polysaccharide essential for many fungi.
- *Pharmacokinetics:* Echinocandins are only available as IV formulations. Half-life range from 11-48 hours and may require loading dose in most treatment regimens.
- *Adverse drug reactions:* Echinocandin is well tolerated, with minor gastrointestinal side effects, and flushing reported. Histamine-mediated symptoms of rasl r.

What feature of fungal cell membranes offers a means for drug selectivity?

Flucytosine

- *Mechanism of action:* Flucytosine penetrates the fungal wall and is converted to flurouracil. Flurouracil competes with uracil and interferes with fungal DNA and RNA synthesis. Flucytosine are therefore fungistatic. Resistance develops from monotherapy and altered metabolism for Flucytosine.
- *Pharmacokinetics:* Flucytosine is effective orally and is distributed to most body tissues, including the CNS. It is eliminated intact in the urine, so the dose must be reduced in patients with renal impairment.
- *Adverse drug reactions:* Dose-dependent myelosuppression (anemia, neutropenia, and thrombocytopenia) can occur. This is due to the toxicity developed in rap w due to DNA and RNA damage. Hepatoxicity with increases in bilirubin and hypoglycemia can also occur.

What is the difference between distribution of amphotericin and flucytosine in the body?

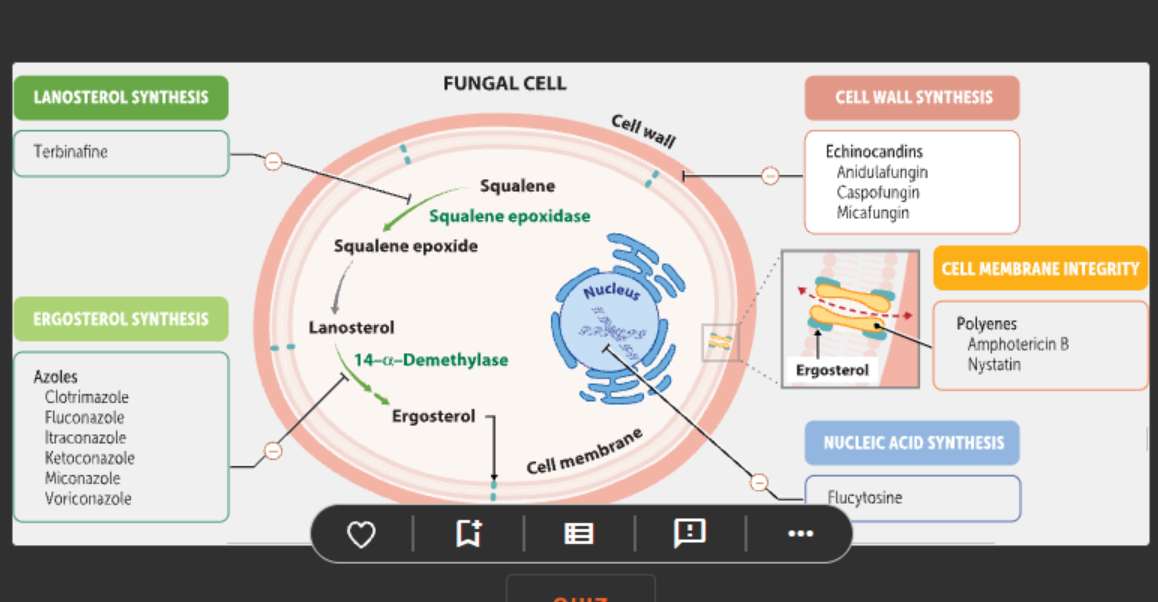
Griseofulvin and Terbinafine

- *Mechanism of action:* Griseofulvin interferes with microtubule function in fungal cells in keratin-rich tissues (e.g., skin, nail, and hair), thereby inhibiting mitosis and fungal cell replication. It acts as a fungistatic. Terbinafine blocks ergosterol synthesis by inhibiting the fungal enzyme squalene epoxidase and forming squalene epoxide from squalene; hence, it acts as a fungicidal. They are available in oral formulations, but given their mechanism of action, they are also available as topical formulations.
- *Adverse effects:* Griseofulvin is contraindicated for pregnancy and severe liver disease. Phe ulvin's use. It is a CYP inducer and can increase the metabolism of hormonal contraceptives (estrogen and progestin), leading to contraceptive failure. Terbinafine effects include GI upsets, rash, headache, and taste disturbances. Hepatotoxicity can also occur with Terbinafine use.

Nystatin

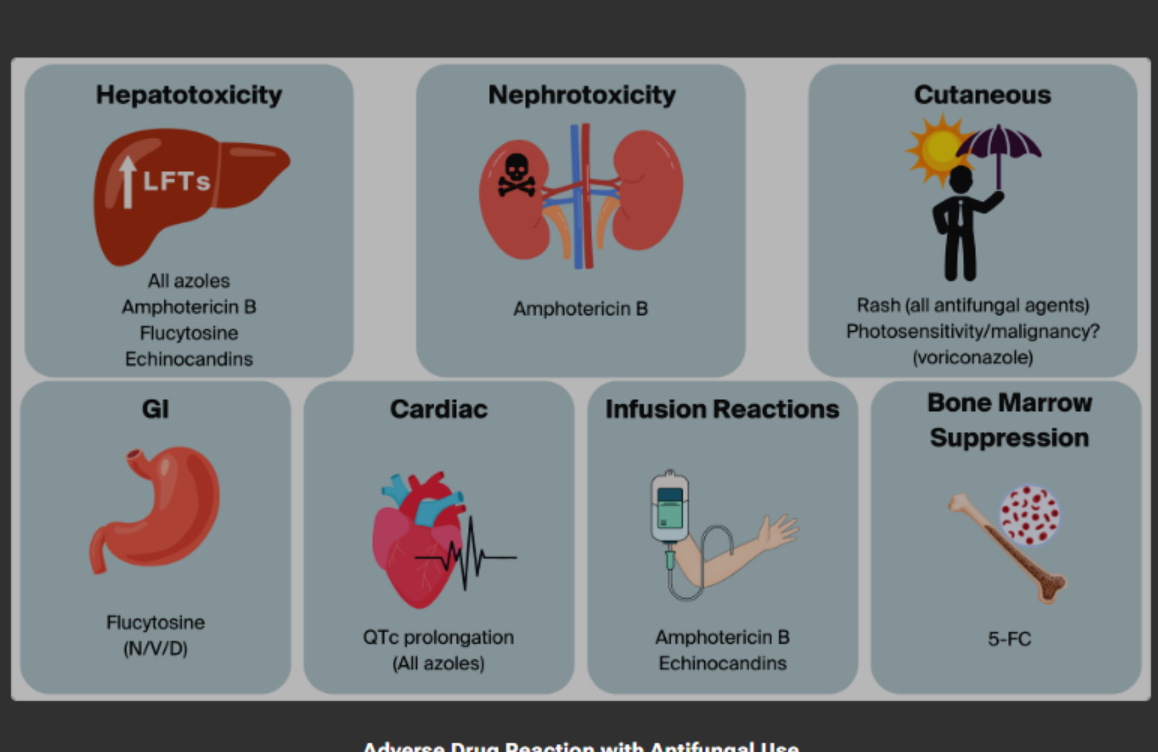
- Mechanism of action: Nystatin is a polyene like Amphotericin B. However, it is toxic for parenteral administrations and is formulated as creams, suspensions (oral use) and suppositories. It is a fungicidal.
- *Spectrum of activity:* The activity is against most Candida species and used in the treatment of oral thrush and vaginal candidiasis.
- *Adverse effects:* Nausea, vomiting, diarrhea (N/V/D) are common. Some electrolytes changes in Mg and K can occur.

Assess you knowledge of the mechanisms of actions !



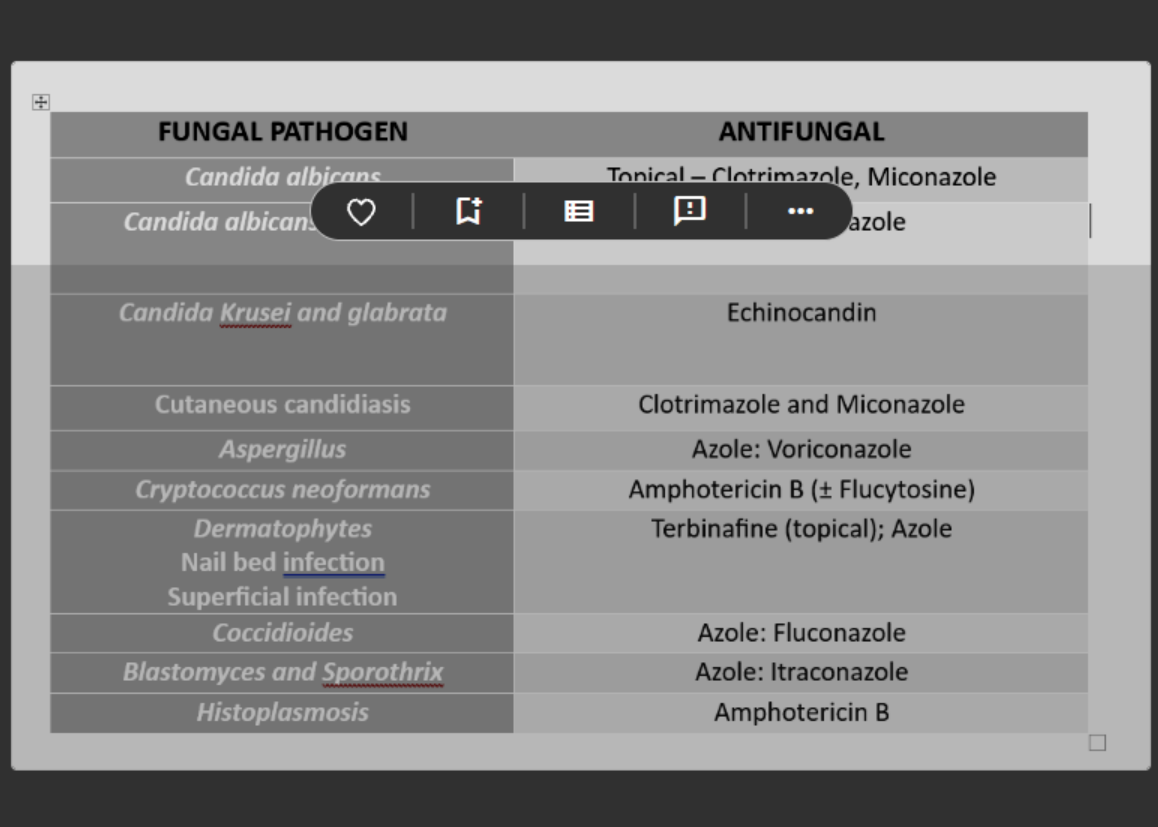
QUIZ

Figure 3



Adverse Drug Reaction with Antifungal Use

What is the spectrum of activity of these Anti-fungal agents? (LO#2)



FUNGAL PATHOGEN	ANTIFUNGAL
Candida albicans	Topical – Clotrimazole, Miconazole
Candida albican	azole
Candida Krusei and glabrata	Echinocandin
Cutaneous candidiasis	Clotrimazole and Miconazole
Aspergillus	Azole: Voriconazole
Cryptococcus neoformans	Amphotericin B (± Flucytosine)
Dermatophytes Nail bed infection Superficial infection	Terbinafine (topical); Azole
Coccidioides	Azole: Fluconazole
Blastomyces and Sporothrix	Azole: Itraconazole
Histoplasmosis	Amphotericin B

Which antifungal drugs affect CYP450 enzymes?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to PL, should she believe her classmates about the worms?

You explain what you need to do, "I think you have a skin condition called ringworm, not just 'worms.' It's pretty common and pretty contagious. To confirm the diagnosis, I'd like to gently scrape off some of those scaly flakes and look at them under my microscope. Assuming this is ringworm, we can give you a medicine, griseofulvin, that will treat it. This is one you need to take by mouth, every day, for about 6 weeks. I know it seems funny to put medicine in your mouth to treat the back of your head, but that's how this one works!" You and PL's mother discuss the treatment plan, prognosis, and details about how it may be caught from infected people or pets. PL tells you, "It's gross to hear I have worms. Don't tell anybody!"

Summary

- Antifungal therapy presents unique challenges due to the similarities between human and fungal cells, which can lead to significant host toxicity.
- Antifungal agents target fungal pathogens selectively, often by interacting with or inhibiting the synthesis of fungal cell walls and ergosterol, a component unique to fungal cell membranes.
- Antifungal agents include polyenes, azoles, echinocandins, griseofulvin and terbinafine, and flucytosine, each with distinct mechanisms of action, resistance patterns, pharmacokinetic properties, and associated adverse drug reactions.
- Polyenes, such as Amphotericin B, bind to ergosterol and cause formation of pores in the ce eath. Azoles inhibit the formation of ergosterol, inhibiting fungal growth.
- Echinocandins inhibit β-glucan synthesis, disrupting the fungal cell wall. Griseofulvin interferes with microtubule function in fungal cells, inhibiting mitosis and fungal cell replication. Terbinafine blocks ergosterol synthesis by inhibiting the fungal enzyme squalene epoxidase.
- Flucytosine interferes with fungal DNA and RNA synthesis. Each of these drugs has specific pharmacokinetic properties, spectrums of activity, and potential adverse drug reactions.

Review Questions

1. Which drug interacts with cell membrane components, resulting in the formation of pores that allow leakage of electrolytes?

A. ○ Caspofungin
B. ○ Flucytosine
C. ○ Griseofulvin
D. ● Nystatin
E. ○ Terbinafine

Hide Explanation

The correct answer i , is a polyene antibiotic that interacts with ergosterol in fungal cell membranes to form artificial pores. Caspofungin (A) inhibits cell wall synthesis. Flucytosine (B) inhibits DNA and RNA biosynthesis. Griseofulvin (C) interferes with microtubule function. Terbinafine (E) inhibits the fungal enzyme squalene epoxidase.

2. Which mechanism of action is associated with Amphotericin B?

A. ○ Inhibiting DNA synthesis
B. ○ Binding to β-glucan
C. ● Binding to ergosterol
D. ○ Inhibiting microtubule function

Hide Explanation

The correct answer is binding to ergosterol (C). Amphotericin B binds to ergosterol in the fungal cell membrane, forming pores that lead to cell death. Inhibiting DNA synthesis (A) is a mechanism of Flucytosine. Binding to β-glucan (C) is a mechanism of echinocandins. Inhibiting microtubule function (E) is a mechanism of griseofulvin.

3. Which of the following topical antifungal is used to treat a fungal nail infection?

A. ○ Caspofur
B. ● Terbinafine
C. ○ Amphotericin B
D. ○ Flucytosine

Hide Explanation

The correct answer is Terbinafine (B). Terbinafine covers dermatophytes and is available as a topical formulation. Caspofungin (A), Amphotericin (B) are IV formulation, and Flucytosine (D) oral and not used as a monotherapy.

4. Which antifungal is contraindicated in pregnancy?

A. ● Griseofulvin
B. ○ Terbinafine
C. ○ Nystatin
D. ○ Amphotericin B
E. ○ Fluconazole

Hide Explanation

The correct answer is Griseofulvin (A). Griseofulvin is contraindicated for pregnancy due to its potential teratogenic effects. Terbinafine (B) and Nystatin (C) do not have this specific contraindication. Amphotericin B (D) and Fluconazole (E) are not mentioned in the context provided.

5. Which of the following would be best to treat a systemic Candidiasis infection?

A. ○ Griseofulvin
B. ○ Terbinafine
C. ○ Nystatin
D. ● Fluconazole

Hide Explanation

The correct answer is Fluconazole (D). Systemic Fluconazole covers systemic Candida infections. Griseofulvin (A) and Terbinafine (B) do not cover Candida infections and are topical formulations. Nystatin (C) is not administered as a parenteral formulation because it causes significant toxicity.

6. Ketoconazole is a strong inhibitor of which CYP isoform?

A. ● CYP 3A4
B. ○ CYP2E1
C. ○ CYP2C19
D. ○ CYP2D6

Hide Explanation

The correct answer is CYP 3A4 (A). Azoles are strong inhibitors of CYP 3A4. They can precipitate serious drug-drug interactions with CYP 3A4 substrates. Azoles are not strong inhibitors of CYP 2E1 (B), CYP 2C19 (C) or CYP 2D6 (D)

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☐ Describe the mechanism of action and mechanism of resistance of antifungal drugs.

02 ☐ List the e treatment of

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Objective Confidence: 0/4